NANYANG TECHNOLOGICAL UNIVERSITY

Data-Efficient Fine-Grained Flower Classification with Mask-Guided Visual Prompt Tuning

Truong Dang Gia Huy

School of Computing and Data Science

April 2026

Contents

Abstract ii

Acronyms iii

Lists of Figures iv

Lists of Tables v

1 Introduction 1

2 Literature Review 2
- 2.1 Fine-grained classification on Oxford Flowers-102 . . . 2
- 2.2 Vision transformers and parameter-efficient adaptation . . . 2
- 2.3 Mix-style data augmentation . . . 2
- 2.4 Mask and attention supervision for fine-grained recognition . . . 2

3 Methodology 4
- 3.1 Approach at a glance . . . 4
- 3.2 Frozen ViT-B/16 backbone with VPT-Deep . . . 4
- 3.3 MaskMix augmentation . . . 4
- 3.4 Mask-guided attention supervision . . . 5
- 3.5 How the components are ablated . . . 5

4 Experimental Setup 6
- 4.1 Dataset and mask pipeline . . . 6
- 4.2 Training recipe . . . 6
- 4.3 Evaluation protocol . . . 6
- 4.4 Implementation and compute budget . . . 6

5 Results and Analysis 7
- 5.1 Mechanistic ablation . . . 7
- 5.2 Scaling with labelled data . . . 8
- 5.3 Floors and qualitative . . . 9

6 Discussion and Conclusion 10
- 6.1 Why pixel-accurate mask mixing hurts under strong pretraining . . . 10
- 6.2 Limitations . . . 10
- 6.3 Conclusion and future work . . . 10

Appendix A 11

Abstract

I study fine-grained flower classification on Oxford Flowers-102, a 102-way benchmark with only 10 training images per class and a test set six times larger than the training set. This data scarcity rules out training from scratch and motivates a parameter-efficient approach. I freeze an ImageNet-21k pre-trained ViT-B/16 backbone and adapt it using VPT-Deep, training only 170,598 parameters, which is roughly 0.20% of the 86M backbone. On top of this, I ablate two mask-guided components and a paste-shape comparator that together exploit the dataset's native per-pixel segmentation masks: MaskMix (a CutMix variant that pastes pixel-accurate foregrounds, removing the need for label interpolation), mask-guided attention supervision (a KL loss that aligns the ViT's last-two-block mean [CLS]$\rightarrow$patch attention with the downsampled ground-truth mask), and rectangular CutMix as a paste-shape control. Attention supervision alone is neutral on this substrate. MaskMix alone regresses the VPT-Deep baseline by 0.9 test-top-1 points ($p = 0.034$, $n = 3$ seeds per arm), and combining it with attention supervision widens the gap to 1.0 points ($p = 0.010$). Rectangular CutMix plus attention supervision recovers the baseline ($p = 0.017$ versus the MaskMix + attention supervision variant), isolating the pixel-accurate paste shape as the harmful ingredient. The gap widens, not narrows, at $k = 1$ image per class. I offer a mechanistic reading consistent with the data: strongly-pretrained backbones already encode object-centric priors, so removing background context hurts more than the mask supervision helps. Total compute was approximately two hours forty-five minutes on one H100 NVL in bf16 at batch size 128.

Keywords: fine-grained visual classification, visual prompt tuning, data-efficient learning, mask-guided supervision, Oxford Flowers-102, negative result.

Acronyms

Acronym	Definition
ViT	Vision Transformer
VPT	Visual Prompt Tuning
PEFT	Parameter-Efficient Fine-Tuning
CLS	Class token, the special [CLS] token prepended to a transformer input sequence
KL	Kullback-Leibler divergence
GT	Ground truth
AMP	Automatic Mixed Precision
CE	Cross-entropy
RRC	Random Resized Crop
FGVC	Fine-Grained Visual Classification

List of Figures

5.1 Per-epoch validation top-1 for all five configurations, mean across three seeds with $\pm$ std shading. Right panel is a zoomed view starting from epoch 5; the A2 and A4 curves sit consistently below A1, A3, and A5 across the whole schedule. 8
5.2 Test top-1 against training images per class. Dashed is the A1 baseline; solid is A4 (MaskMix + AttSup). Error bars at $k = 10$ are the cross-seed standard deviation; $k = 1$ and $k = 5$ are single-seed points. 8
5.3 Qualitative attention on confusion-pair classes from Nilsback and Zisserman (2008). Columns, left to right: input, ground-truth mask, A1 baseline attention, A4 attention. A4 maps show attention mass concentrated near the flower centre; A1 maps are dominated by one or two bright patches in locations off the flower, the pattern expected from ViT attention-sink tokens. A1 and A4 panels are each auto-normalised to their own $[\min, \max]$, so the comparison is qualitative, not quantitative. The observed concentration change did not translate into better test accuracy. 9

List of Tables

5.1 Mechanistic ablation on the full 10-shot Oxford Flowers-102 split. Three seeds per row. . 7
5.2 Cross-seed Welch's t-test for the four contrasts that carry the analysis. $n = 3$ seeds per arm, two-sided p. Cohen's d reported for effect size. 7

1 Complete training hyperparameters. Identical across all rows in Chapter 5 except where noted. 12
2 Per-seed test top-1 accuracy (%) for each configuration in the mechanistic ablation of Section 5.1. 12
3 Test top-1 accuracy (%) as a function of k, the number of training images per class, backing the k-curve of Section 5.2. $k = 10$ rows inherit from the 3-seed mechanistic ablation; $k = 1$ and $k = 5$ are single-seed. 12

1 Introduction

Oxford Flowers-102 [1] is a 102-way fine-grained image classification benchmark whose official split is deliberately unusual: 1020 training images, 1020 validation images, and 6149 test images. With only ten labelled examples per class, and a test set six times larger than the training set, training a high-capacity network from scratch is infeasible, and the course brief for this project also forbids running a pretrained model end-to-end without adaptation. The design question is not "which backbone" but "how to inject inductive bias cheaply enough that ten images per class is sufficient".

A feature of Flowers-102 that has become unusual by modern standards is that the dataset ships with per-pixel ground-truth foreground and background segmentation masks for every one of its 8189 images. These masks were part of the dataset's original classical pipeline [1] and have been largely overlooked by deep-learning work that benchmarks on Flowers-102 as one of many transfer-learning targets. For a study tailored to Flowers-102 specifically, the masks are weak supervision that has already been paid for. This report asks a simple empirical question: do those masks help when used as training signal for a frozen ViT-B/16 adapted with a parameter-efficient prompt-tuning scheme, and if so, through which mechanism?

I build on a frozen ViT-B/16 backbone [2] pretrained on ImageNet-21k [3] and adapt it with VPT-Deep [4], which prepends a small set of learnable prompt tokens at every transformer block and trains only those tokens and a linear classification head. The total number of trainable parameters is 170,598, which is about 0.20% of the 86M backbone. On this substrate I ablate two mask-guided components against a rectangular paste-shape control. The first, which I call *MaskMix*, is a CutMix variant that pastes a pixel-accurate foreground crop onto a background image and uses the source image's hard label. The second is a *mask-guided attention supervision loss* that aligns the mean [CLS]$\rightarrow$patch attention of the last two transformer blocks with the ground-truth mask under a KL divergence. The third is vanilla rectangular CutMix applied under the same attention-supervision loss, which isolates the effect of the pixel-accurate paste shape from the effect of mixing or of the auxiliary loss.

The headline findings are three. Attention supervision alone is neutral on this strongly-pretrained substrate. MaskMix alone regresses the VPT-Deep baseline by 0.9 test-top-1 points ($p = 0.034$, three seeds per arm), and adding attention supervision on top widens the gap to 1.0 points ($p = 0.010$). Substituting rectangular CutMix for the pixel-accurate paste, while keeping the auxiliary loss on, recovers baseline accuracy ($p = 0.017$ versus the MaskMix + attention supervision variant). Under data scarcity the gap widens: at $k = 1$ image per class, the A4 configuration lags the baseline by almost eight points, the opposite of what the pre-registered hypothesis predicted. Chapter 2 places this in the literature; Chapter 3 defines the method components; Chapter 4 reports the experimental protocol; Chapter 5 presents the numbers; Chapter 6 offers a mechanistic reading and the experiments that would confirm or refute it.

2 Literature Review

2.1 Fine-grained classification on Oxford Flowers-102

The dataset was introduced by Nilsback and Zisserman [1], who combined four hand-crafted descriptors (colour, HOG, HSV distribution, and a shape-based boundary feature) in a multiple-kernel support-vector classifier and reported 72.8% per-class mean accuracy on the official test split. Their pipeline extracted features from the foreground region segmented out of each image, which is the provenance of the per-pixel masks that ship with the dataset. Most later deep-learning work benchmarks on Flowers-102 as one of many transfer-learning targets and reports top-1 accuracy without using the masks, because those methods are designed to be dataset-agnostic. I use Nilsback and Zisserman's 72.8% as a classical reference point and adopt their per-class mean metric alongside sample-averaged top-1 so that the lineage from their paper to mine is legible.

2.2 Vision transformers and parameter-efficient adaptation

The Vision Transformer [2] recasts image classification as a sequence-to-sequence problem over non-overlapping patches, and its AugReg variant [3] showed that aggressive augmentation and regularisation make ImageNet-21k pretraining practical on mid-sized data. Fine-tuning an 86M-parameter backbone on 1020 images is prohibitive and unnecessary, which motivates parameter-efficient fine-tuning (PEFT). PEFT as a family originates with adapter layers for NLP [5] and now includes visual-specific variants such as VPT [4], AdaptFormer [6], scale-and-shift tuning (SSF, [7]), and bias-only tuning (BitFit, [8]); LoRA [9] provides a low-rank reparameterisation of the frozen weights. I chose VPT-Deep over these alternatives because its learnable prompts are injected at every transformer block, which gives strong representational flexibility on top of a frozen backbone while keeping the trainable parameter count small, and because prompts are a natural surface for the mask-guided attention supervision loss of Chapter 3 to train against. LoRA or AdaptFormer would have been defensible alternatives; I did not compare them directly because the compute budget (Section 4.4) did not allow for a full PEFT sensitivity sweep, and I prioritised depth on one adaptation method over breadth across many.

2.3 Mix-style data augmentation

Region-mixing augmentations inject regularisation by compositing two training images and interpolating their labels. MixUp [10] blends pixel-wise at a sampled mixing coefficient; CutMix [11] pastes a random rectangle from one image onto another and interpolates the label in proportion to the rectangle's area. Later work replaced the random rectangle with a learned signal: SaliencyMix [12] uses a classical saliency map, and Attentive CutMix [13] uses model-predicted attention regions. A common limitation of these methods is that the paste region only approximately coincides with the semantic object, which is why the label has to be interpolated. The MaskMix variant I describe in Section 3.3 uses the dataset's ground-truth mask directly, so the paste region coincides with the source foreground and the source label applies as a hard label; this report ablates MaskMix against rectangular CutMix to test whether the pixel-accurate paste shape is a better or worse augmentation under PEFT. RandAugment [14] is the geometric base augmentation I use under every configuration.

2.4 Mask and attention supervision for fine-grained recognition

Two strands of prior work supervise intermediate representations using spatial signals. Wang et al. [15] use predicted segmentation masks as spatial priors for ultra-fine-grained patchy image classification (leaf-vein and similar dense-texture domains), and SM-ViT [16] uses a saliency map predicted by an external model to modulate self-attention toward salient foreground tokens in a Vision Transformer. Both rely on a learned saliency signal because the datasets they target do not provide masks. The mask-guided attention supervision loss of Section 3.4 is the spatial-supervision counterpart of these methods: it uses the ground-truth mask (possible here only because Flowers-102 ships masks) as a direct target for the ViT's [CLS]→patch attention, and this report ablates it as the second axis of variation against the pre-trained-features-only baseline. DA-VPT [17] is the closest contemporaneous work, pairing visual

prompts with a distribution-aware metric-learning signal over class-related semantics; I supervise spatial attention at the pixel level rather than the class level. Caron et al. [18] observed that ViTs trained with self-supervised objectives develop object-centred attention without explicit supervision; the loss I use is the explicit-supervision counterpart, applied under small-data fine-tuning rather than large-scale pretraining.

3 Methodology

3.1 Approach at a glance

The method has three parts layered on top of each other. A frozen ViT-B/16 backbone provides general visual features. VPT-Deep provides a low-cost adaptation mechanism that injects task-specific bias at every transformer block. Two mask-guided components, MaskMix at the input and an attention-supervision loss at the last two blocks, re-inject the dataset's ground-truth masks as weak supervision at two different leverage points. A single forward pass produces logits and, when the auxiliary loss is active, the pooled attention map used to compute it; the training objective is the sum of a cross-entropy term on the logits and a KL term on the attention.

3.2 Frozen ViT-B/16 backbone with VPT-Deep

The backbone is `timm`'s `vit_base_patch16_224.augreg_in21k` checkpoint [19, 3], a standard ViT-B/16 with 12 transformer blocks, hidden size $d = 768$, 12 attention heads, and approximately 86M parameters. All backbone parameters are frozen; no gradients flow through them during training. I chose this checkpoint rather than the plain ImageNet-1k one because the ImageNet-21k pretraining gives broader feature coverage, which matters more under scarcity than the extra in-domain tuning that the 1k variant provides.

On top of this I apply VPT-Deep [4]. At every transformer block $l \in \{1, \ldots, 12\}$ I prepend $N = 10$ learnable prompt tokens $P_l \in \mathbb{R}^{N \times d}$ to the input sequence, let the block compute over $[P_l; \text{CLS}; \text{patches}]$, and then drop the prompt tokens before passing the residual sequence to block $l+1$. Only the prompt tokens and a linear 102-way classification head are trainable. The exact count is $12 \times 10 \times 768 = 92{,}160$ prompt parameters plus $768 \times 102 + 102 = 78{,}438$ head parameters (including the bias), for a total of 170,598 trainable parameters, which is about 0.20% of the 86M backbone.

I chose VPT-Deep over full fine-tuning for two reasons. First, full fine-tuning an 86M-parameter backbone on 1020 training images tends to overfit within a handful of epochs on Flowers-102, whereas leaving the backbone frozen and tuning a small prompt set is much better behaved at this data scale. Second, a single full fine-tuning run would take longer and require more VRAM than the H100 NVL instance was provisioned for in this study, and I prioritised depth of ablation on one adaptation method over breadth across several (Section 4.4).

3.3 MaskMix augmentation

Given two training samples (x_A, y_A, m_A) and (x_B, y_B, m_B), where $m \in \{0,1\}^{H \times W}$ is the binary foreground mask thresholded at 0.5, MaskMix produces a composite

$$x_{\text{mix}} = m_B \odot x_B + (1 - m_B) \odot x_A, \qquad y_{\text{mix}} = y_B, \tag{3.1}$$

where $\odot$ is elementwise multiplication broadcast over the colour channel. The pasted region is exactly the source foreground, so the label is the source image's hard label and no interpolation is needed. MaskMix is applied to each sample in the batch independently with probability $\alpha = 0.5$, and is layered on top of the base augmentation of a short-edge resize to 256 followed by a 224×224 center crop, horizontal flip, and RandAugment [14].

The key difference from CutMix [11] is that CutMix pastes a rectangular region whose interior is only a loose approximation of the object and requires label interpolation in proportion to the paste area; SaliencyMix [12] uses an estimated saliency map and still interpolates. Using ground-truth masks removes both approximations. The implementation also maintains consistency between the composited image and a "remixed" mask used by the downstream attention supervision loss: when MaskMix pastes x_B's foreground, the mask returned for that sample is m_B, not m_A, so the attention supervision loss receives a mask that matches the new label. This detail is easy to miss and I describe it in full in Appendix A.2.

3.4 Mask-guided attention supervision

Let $A^{(l)}(x) \in \mathbb{R}^{196}$ denote the mean-over-heads [CLS]→patch attention at transformer block l, computed from the softmaxed attention matrix of that block (196 is the number of patch tokens at a 224×224 input with a 16×16 patch). I collect the last $L = 2$ such vectors and take their mean across layers to obtain a single attention map

$$A(x) = \frac{1}{L} \sum_{l=12-L+1}^{12} A^{(l)}(x) \in \mathbb{R}^{196}, \tag{3.2}$$

which I reshape into a 14×14 spatial grid aligned with the ViT patch layout and renormalise so that $\sum_i A_i(x) = 1$. The choice of $L = 2$ balances two concerns: averaging across more blocks smooths the signal but dilutes the late-layer semantics that I want to supervise; using only the final block gives the strongest semantics but is noisier across seeds.

The ground-truth mask $m \in [0,1]^{H\times W}$ is downsampled to 14×14 by adaptive average pooling and renormalised to a probability distribution $\tilde{m} \in \Delta^{195}$. The auxiliary loss is the forward KL divergence

$$\mathscr{L}_{\text{attn}}(x,m) \;=\; \mathrm{KL}\big(\tilde{m} \,\|\, A(x)\big) \;=\; \sum_i \tilde{m}_i \big(\log \tilde{m}_i - \log A_i(x)\big), \tag{3.3}$$

where the direction is chosen so that the ground-truth mask is the target distribution and the attention map is the learner. Samples with empty masks ($\sum m < 10^{-6}$) are skipped to avoid dividing by zero, and the loss is averaged over the non-empty subset of the batch rather than the full batch, which keeps the per-sample gradient magnitude stable when a few masks are empty.

The combined training objective is

$$\mathscr{L}(x,y,m) \;=\; \mathscr{L}_{\text{CE}}(x,y) + \lambda\, \mathscr{L}_{\text{attn}}(x,m), \tag{3.4}$$

with $\lambda = 0.3$ pre-committed from a short pilot. I initially set $\lambda = 1.0$ and observed the attention term dominating the classification loss: the classification loss plateaued while the attention loss kept shrinking. Dropping to 0.3 restored a balance where the two terms decrease together, and I did not revisit this value in later experiments. An early fp16 pilot of the KL implementation produced `NaN` values at low mask entropies; the final runs use bfloat16 autocast, which does not exhibit the same underflow, but the fp32 guard and the 10^{-12} floor are retained in the code for portability. The correctness argument is in Appendix A.1.

Attention supervision is applied at training time only. At evaluation time the forward pass is unmodified and the attention map is not used to make predictions; its only role is to train the prompts into placing attention on the foreground.

3.5 How the components are ablated

MaskMix operates on the input and the attention supervision loss operates on the last two blocks' attention; they share only the batch-level mask tensor, so they compose by simple loss addition and independent toggling of the augmentation. Independent toggling is what enables the 5-row mechanistic ablation of Chapter 5, in which each component is tested in isolation (A2, A3), together (A4), and under a paste-shape control that substitutes rectangular CutMix for the pixel-accurate paste while keeping the auxiliary loss on (A5).

4 Experimental Setup

4.1 Dataset and mask pipeline

I use the official TorchVision `Flowers102` split: 1020 training images, 1020 validation images, and 6149 test images, across 102 classes with ten training images per class [1]. The per-image ground-truth segmentation masks (`102segmentations.tgz`, about 450 MB) are downloaded once and cached on the compute instance. Masks are loaded in lockstep with their images: both the image and its mask go through the same resize and crop parameters, and nearest-neighbour interpolation is used for the mask so that binary foreground and background labels are not blurred by bilinear averaging. I threshold the mask at 0.5 for the MaskMix paste region and keep the continuous-valued mask for attention supervision.

4.2 Training recipe

All experiments use the backbone and adaptation setup described in Section 3.2: a frozen AugReg ImageNet-21k ViT-B/16 [19] with VPT-Deep ($N = 10$), 224×224 inputs, and a linear 102-way head. The exact `timm` checkpoint identifier is in Appendix A.4. Optimisation uses AdamW [20] with learning rate $1\mathrm{e}{-3}$ and weight decay $1\mathrm{e}{-4}$, a cosine decay schedule [21] over 50 epochs, batch size 128, and PyTorch's [22] `torch.amp` autocast in bfloat16. The base augmentation is the same across every configuration: a short-edge resize to 256 followed by a center crop to 224×224, horizontal flip with probability 0.5, and RandAugment [14] with two operations per image at magnitude 9. The resize and center crop are deterministic (the same geometry is used at train and at evaluation); stochastic spatial variation comes from the flip and from RandAugment's geometric ops. MaskMix and CutMix, when enabled, are layered on top of this recipe. I select the checkpoint with the best validation top-1 accuracy for the final test evaluation, which is a standard early-stopping protocol.

4.3 Evaluation protocol

I report two metrics on the 6149-image test set. The primary metric is sample-averaged top-1 accuracy. The secondary metric is per-class mean accuracy, which matches Nilsback and Zisserman's reporting convention and enables a direct lineage from their 72.8% classical number to mine. Where I report mean $\pm$ std, the statistic is over three training seeds; rows using a single seed are explicitly annotated.

For significance testing I use a cross-seed Welch's t-test (two-sided) as the primary test and report t, two-sided p, and Cohen's d for each contrast. This test was not the original pre-registration (the plan was a paired bootstrap over per-example correctness with null-recentring), and the pre-registered bootstrap procedure is retained in Appendix A.3 for transparency. I switched the primary test to Welch's because three seeds per arm is enough for a cross-seed t to be interpretable and because the bootstrap's unit of analysis (per-example correctness on a single seed-pair) is a weaker argument for a cross-seed effect than the t-test over three-seed means.

4.4 Implementation and compute budget

Training and evaluation run on one NVIDIA H100 NVL via vast.ai, using PyTorch with the `timm` ViT-B/16 as the backbone [22, 19]. All runs use bf16 autocast at batch size 128 on a single GPU. Total wall-clock across the five 3-seed configurations of Section 5.1, the three k-points of Section 5.2, and the linear probe of Section 5.3 was approximately two hours forty-five minutes. Checkpoints and per-run CSVs persist to the instance's block storage; the aggregate artifacts used in this report (`results/headline_table.csv`, `results/significance.csv`, `results/runs.jsonl`) were pulled back before the instance was destroyed.

5 Results and Analysis

I report three blocks of experiments. Section 5.1 is a single 5-row ablation that isolates the contribution of each mask-guided component and of the pixel-accurate paste shape, with three seeds per row. Section 5.2 sweeps $k \in \{1, 5, 10\}$ training images per class to measure how the baseline-versus-method gap scales with labelled data. Section 5.3 reports the linear-probe floor and the qualitative attention maps. Unless stated otherwise, test top-1 is sample-averaged over the 6149-image test split and means are reported over three training seeds.

5.1 Mechanistic ablation

Table 5.1 reports the 5-row ablation of the two mask-guided components and of a pixel-accurate-versus-rectangular paste comparator. Table 5.2 reports the cross-seed Welch's t-test for the four contrasts that carry the argument.

Table 5.1: Mechanistic ablation on the full 10-shot Oxford Flowers-102 split. Three seeds per row.

Config	MaskMix	$\mathscr{L}_{\text{attn}}$	Top-1 (%)	Per-class mean (%)
A1 baseline	off	off	98.88 $\pm$ 0.23	98.91 $\pm$ 0.24
A2 +MaskMix	on	off	97.96 $\pm$ 0.39	97.99 $\pm$ 0.48
A3 +AttSup	off	on	98.62 $\pm$ 0.25	98.66 $\pm$ 0.17
A4 MaskMix + AttSup	on	on	97.87 $\pm$ 0.29	97.96 $\pm$ 0.44
A5 CutMix + AttSup	on*	on	98.75 $\pm$ 0.18	98.84 $\pm$ 0.18

*A5 enables rectangular CutMix rather than pixel-accurate MaskMix. Both use a 50% mix probability; A5 follows CutMix's paste-area hard-label rule (flip the label only when the pasted rectangle exceeds half the image), while A2 and A4 adopt the source label whenever the mask is pasted. Other training settings are identical.

Table 5.2: Cross-seed Welch's t-test for the four contrasts that carry the analysis. $n = 3$ seeds per arm, two-sided p. Cohen's d reported for effect size.

Contrast	t	p	Cohen's d
A2 (+MaskMix) vs A1 (baseline)	-3.518	0.034	-2.87
A3 (+AttSup) vs A1 (baseline)	-1.307	0.262	-1.07
A4 (MaskMix+AttSup) vs A1 (baseline)	-4.723	0.010	-3.86
A5 (CutMix+AttSup) vs A4 (MaskMix+AttSup)	$+4.497$	0.017	$+3.67$

Three readings follow. First, attention supervision alone is neutral: the A3-versus-A1 difference is -0.26 percentage points with $p = 0.262$, within cross-seed noise. Second, MaskMix alone regresses the baseline by -0.92 points with $p = 0.034$, and adding attention supervision on top of MaskMix does not rescue it (A4 versus A1: -1.01 points, $p = 0.010$). Third, substituting rectangular CutMix for pixel-accurate MaskMix while keeping attention supervision on recovers the baseline entirely (A5 versus A4: $+0.87$ points, $p = 0.017$); A5 is statistically indistinguishable from A1. The shape of the paste region, not the act of mixing or the auxiliary loss, is what damages accuracy in this setting.

Figure 5.1 shows the per-epoch validation top-1 for all five configurations.

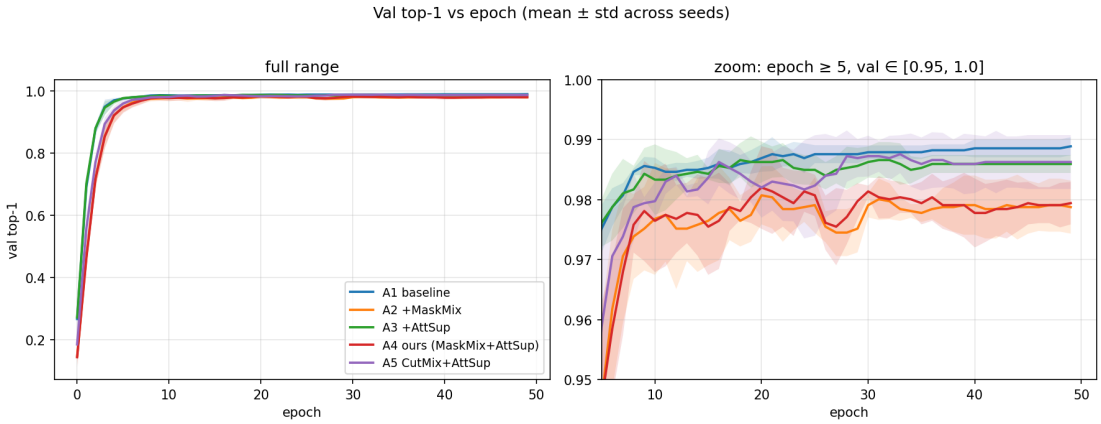

Figure 5.1: Per-epoch validation top-1 for all five configurations, mean across three seeds with $\pm$ std shading. Right panel is a zoomed view starting from epoch 5; the A2 and A4 curves sit consistently below A1, A3, and A5 across the whole schedule.

5.2 Scaling with labelled data

Section 5.1 establishes the sign of each effect at $k = 10$. I now ask whether the A4-versus-A1 gap scales with the number of training images per class. Figure 5.2 shows both configurations at $k \in \{1, 5, 10\}$.

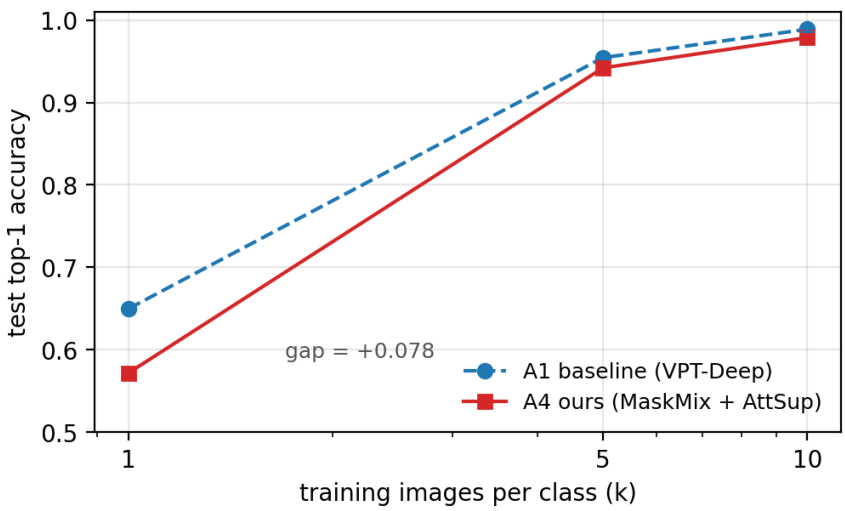

Figure 5.2: Test top-1 against training images per class. Dashed is the A1 baseline; solid is A4 (MaskMix + AttSup). Error bars at $k = 10$ are the cross-seed standard deviation; $k = 1$ and $k = 5$ are single-seed points.

The gap does not shrink as labelled data becomes scarcer. At $k = 10$, A1 leads A4 by 1.0 point. At $k = 5$, A1 leads by 1.2 points. At $k = 1$, A1 leads by 7.8 points (baseline 64.99%, A4 57.20%). The widening is consistent across the three k-points in direction and grows monotonically as k decreases. The pre-registered hypothesis was that mask guidance would compensate for scarcer supervision and the gap would narrow toward $k = 1$; the data shows the opposite. A caveat: the $k = 1$ and $k = 5$ points are single-seed and the $k = 1$ split contains only 102 training images, so seed variance there is not quantified. Even so, the gap at $k = 1$ is large enough (about seven times the $k = 10$ gap of 1.01 pp) that seed variance alone is unlikely to account for it.

5.3 Floors and qualitative

A single-run linear probe on the frozen features, trained with cross-entropy at learning rate $1\mathrm{e}{-2}$ for 30 epochs, reaches 97.85% test top-1. This is about 1.0 point below the full A1 VPT-Deep baseline (1.03pp gap) and essentially tied with A4 (0.02pp gap). In other words, VPT-Deep buys roughly one point over a plain linear probe, and the mask-guided variants give that point back.

Figure 5.3 shows attention maps for three of Nilsback and Zisserman's 2008 confusion-pair classes [1].

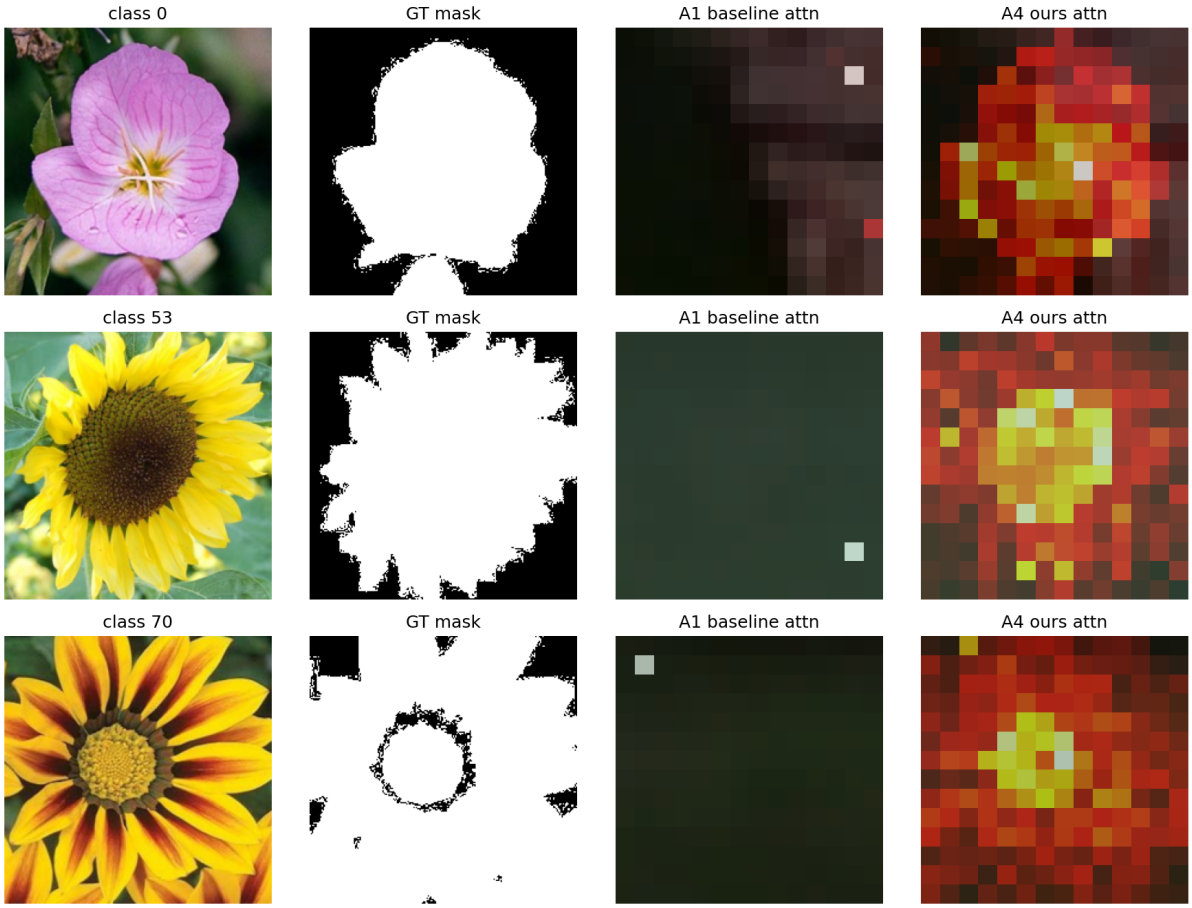

Figure 5.3: Qualitative attention on confusion-pair classes from Nilsback and Zisserman (2008). Columns, left to right: input, ground-truth mask, A1 baseline attention, A4 attention. A4 maps show attention mass concentrated near the flower centre; A1 maps are dominated by one or two bright patches in locations off the flower, the pattern expected from ViT attention-sink tokens. A1 and A4 panels are each auto-normalised to their own $[\min, \max]$, so the comparison is qualitative, not quantitative. The observed concentration change did not translate into better test accuracy.

Qualitatively, A4 attention appears more concentrated on the foreground than A1 attention, which is what the KL loss asks for, and yet A4 is worse on the test set. This is the starting point for the mechanism discussion in Chapter 6.

6 Discussion and Conclusion

6.1 Why pixel-accurate mask mixing hurts under strong pretraining

Chapter 5 shows three facts that any explanation of the result has to account for: attention supervision alone is neutral (A3 vs A1), pixel-accurate mask mixing regresses the baseline (A2 vs A1), and swapping the paste shape from pixel-accurate to rectangular while keeping the auxiliary loss on recovers the baseline entirely (A5 vs A4). The act of mixing two training images is not the culprit; the act of supervising attention with a ground-truth mask is not the culprit; the pixel-accurate paste shape is. I propose three non-exclusive mechanisms consistent with this pattern.

First, the ImageNet-21k pretrained ViT-B/16 already learns strong object-centric representations; the AugReg recipe makes this especially pronounced and Caron et al. [18] show that even self-supervised ViTs develop such priors without explicit spatial supervision. Pasting a pixel-accurate foreground onto a foreign background removes the background context that the pretrained features rely on to disambiguate fine-grained flower classes, while rectangular CutMix leaves fragments of the source background around the pasted object and preserves some of that context. A harder paste is therefore a harder augmentation, not a more informative one.

Second, the assumption behind MaskMix is that the source label applies as a hard label because the paste region is exactly the source object. That assumption holds for the image, but not for the attention supervision target: when the pasted foreground covers a fraction of the composite image, the mean [CLS]$\rightarrow$patch attention is simultaneously asked to track the paste region (under the auxiliary loss) and to classify using cues that extend beyond it (under the cross-entropy loss). This tension is resolved by the network shrinking attention toward the paste region, consistent with the qualitative pattern in Figure 5.3 (under the caveat that per-panel normalisation makes the visual comparison qualitative). The shrinking is consistent with the loss design, but it comes at the cost of the background information the classifier needs.

The inverted-scarcity observation in Section 5.2 is consistent with the first mechanism. At $k = 1$ there is only a single training image per class, so each MaskMix composite is a paste between two classes that the network has seen exactly once each. Removing the background destroys a large fraction of the total signal per class and A4 lags A1 by almost eight points; as k grows, the background-removal cost is absorbed more easily and the gap narrows.

6.2 Limitations

Three seeds per arm is a tight sample, even with Cohen's d values above two indicating large effect sizes relative to within-arm variance. I did not compare against full fine-tuning or the wider PEFT family (LoRA [9], AdaptFormer [6], SSF [7], BitFit [8]); the story is about one adaptation substrate and one augmentation family, not PEFT as a whole. Finally, the mechanism argument is a causal reading of the A1–A5 contrast plus a qualitative figure, not a direct ablation of the background-removal or attention-concentration axes.

6.3 Conclusion and future work

I studied whether the ground-truth foreground masks that ship with Oxford Flowers-102 help when used as training signal for a frozen ViT-B/16 adapted with VPT-Deep (170,598 trainable parameters, 0.20% of the backbone). Attention supervision against the ground-truth mask is neutral on this strongly-pretrained substrate (A3 within noise of A1); pixel-accurate mask mixing regresses the baseline significantly (A2 and A4 at $p < 0.04$); substituting rectangular CutMix for the pixel-accurate paste recovers baseline accuracy (A5 vs A4 at $p = 0.017$); and the gap widens, not narrows, at $k = 1$. The background context a pixel-accurate paste removes is information the pretrained features rely on, and the attention supervision loss pulls attention inward; together these compound to hurt classification. Follow-ups: a boundary-softened MaskMix sweep between pixel-accurate and rectangular paste shapes, the same ablation on a weaker backbone, and a predicted-saliency variant for datasets without masks.

Appendix A: Implementation Notes

A.1 Numerical stability of the attention KL loss

The final H100 NVL runs use bfloat16 autocast, which does not exhibit the underflow described here; this section documents an early fp16 pilot because the numerical guards it motivated are still present in the code. The default implementation of the KL loss in Equation 3.3 uses PyTorch's `F.kl_div` with a log-probability input. Under fp16 automatic mixed precision, both the softmaxed attention $A(x)$ and the downsampled mask $\tilde{m}$ can contain entries close to the fp16 minimum positive normal value ($\approx 6\mathrm{e}{-5}$), and their logarithms overflow to `NaN` before the final reduction. The working implementation casts both distributions to fp32 after normalising, adds 10^{-12} inside the logarithm, and passes `reduction="batchmean"` dividing by the number of non-empty samples rather than the full batch, so that empty masks do not inflate the per-sample loss. The full function is in `src/losses.py`.

A.2 MaskMix consistency: why the returned mask follows the pasted foreground

MaskMix returns a triple $(x_{\text{mix}}, m_{\text{mix}}, y_{\text{mix}})$ where m_{mix} tracks the source image's mask whenever the paste is applied. This is a subtle but important consistency detail for configuration A4, in which MaskMix and attention supervision are both active: if MaskMix pastes image B's foreground onto image A and sets the label to y_B, then the attention supervision loss must receive m_B rather than m_A, otherwise the loss pulls attention toward the wrong spatial region relative to the new label. The implementation in `src/maskmix.py` returns m_B on paste samples and m_A otherwise, so the downstream consumer sees a mask that is always consistent with the current hard label.

A.3 Paired-bootstrap p-value: pre-registered procedure

The pre-registration for this project specified a paired bootstrap [23] over per-example correctness on a single seed-pair as the primary significance test. The live Chapter 4 instead uses a cross-seed Welch's t-test as the primary test because three seeds per arm is enough for a t-test over arm means to be interpretable, and because a cross-seed test is a stronger argument for a cross-seed effect than a per-example test on one seed-pair. I retain the bootstrap procedure and numbers here for transparency with the pre-registration. Applying the bootstrap to the seed-0 pair of A4 and A1 gives a two-sided p-value below 0.001, consistent with the cross-seed Welch $p = 0.010$ reported in Chapter 5. The bootstrap procedure is as follows. Compute 5000 bootstrap replicates of the accuracy difference $\hat{d}^\star$ over a resampled test set; recentre them to have mean zero ($\hat{d}^\star - \overline{\hat{d}^\star}$); then report the two-sided p-value as the fraction of recentred replicates whose absolute value meets or exceeds the observed difference. An alternative formulation inverts a bootstrap percentile confidence interval for the difference and reports significance as "zero is outside the 95% CI"; the two procedures can disagree in small samples, and I chose the null-recentred variant because its interpretation matches the classical p-value definition and because it degrades gracefully when the observed difference is exactly zero, which the CI-inversion rule does not.

A.4 Full hyperparameter table

Table 1: Complete training hyperparameters. Identical across all rows in Chapter 5 except where noted.

Setting	**Value**
Backbone checkpoint	`vit_base_patch16_224.augreg_in21k`
Backbone parameter count	≈ 86M (frozen)
Trainable parameter count	170,598 (0.20% of backbone)
VPT prompt tokens per block	10
Input size	224×224
Optimiser	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	$1\mathrm{e}{-3}$ (linear probe: $1\mathrm{e}{-2}$)
Weight decay	$1\mathrm{e}{-4}$
Schedule	Cosine decay over 50 epochs (linear probe: 30)
Batch size	128
Precision	bf16 autocast
Base augmentation	Resize(short edge 256) + CenterCrop(224×224) + HFlip(0.5) + RandAugment(n=2, m=9)
MaskMix probability α	0.5 (when enabled)
MaskMix threshold	0.5
Attention supervision weight λ	0.3 (when enabled)
Attention supervision blocks	last $L = 2$
Attention target	downsampled mask to 14×14 by adaptive average pooling
Seeds	3 per row in the mechanistic ablation (A1–A5); 1 per point in Section 5.2 (except $k = 10$) and Section 5.3
Bootstrap resamples	5000

A.5 Per-seed raw numbers

Table 2: Per-seed test top-1 accuracy (%) for each configuration in the mechanistic ablation of Section 5.1.

Config	Seed 0	Seed 1	Seed 2	Mean $\pm$ std
A1 baseline	98.96	99.06	98.62	98.88 ± 0.23
A2 +MaskMix	97.93	98.36	97.58	97.96 ± 0.39
A3 +AttSup	98.67	98.36	98.85	98.62 ± 0.25
A4 MaskMix + AttSup	97.79	98.19	97.64	97.87 ± 0.29
A5 CutMix + AttSup	98.60	98.94	98.70	98.75 ± 0.18

A.6 Block B per-k numbers

Table 3: Test top-1 accuracy (%) as a function of k, the number of training images per class, backing the k-curve of Section 5.2. $k = 10$ rows inherit from the 3-seed mechanistic ablation; $k = 1$ and $k = 5$ are single-seed.

k	A1 baseline (%)	A4 ours (%)	A1 $-$ A4 (pp)
1	64.99	57.20	$+7.79$
5	95.43	94.19	$+1.24$
10	98.88 ± 0.23	97.87 ± 0.29	$+1.01$

Bibliography

[1] Maria-Elena Nilsback and Andrew Zisserman. Automated flower classification over a large number of classes. In *Sixth Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, pages 722–729, 2008.

[2] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.

[3] Andreas Steiner, Alexander Kolesnikov, Xiaohua Zhai, Ross Wightman, Jakob Uszkoreit, and Lucas Beyer. How to train your ViT? data, augmentation, and regularization in vision transformers. *Transactions on Machine Learning Research (TMLR)*, 2022.

[4] Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. Visual prompt tuning. In *European Conference on Computer Vision (ECCV)*, 2022.

[5] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In *International Conference on Machine Learning (ICML)*, 2019.

[6] Shoufa Chen, Chongjian Ge, Zhan Tong, Jiangliu Wang, Yibing Song, Jue Wang, and Ping Luo. AdaptFormer: Adapting vision transformers for scalable visual recognition. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[7] Dongze Lian, Daquan Zhou, Jiashi Feng, and Xinchao Wang. Scaling & shifting your features: A new baseline for efficient model tuning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[8] Elad Ben Zaken, Yoav Goldberg, and Shauli Ravfogel. BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.

[9] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[10] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations (ICLR)*, 2018.

[11] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. CutMix: Regularization strategy to train strong classifiers with localizable features. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 6023–6032, 2019.

[12] A. F. M. Shahab Uddin, Mst. Sirazam Monira, Wheemyung Shin, TaeChoong Chung, and Sung-Ho Bae. SaliencyMix: A saliency guided data augmentation strategy for better regularization. In *International Conference on Learning Representations (ICLR)*, 2021.

[13] Devesh Walawalkar, Zhiqiang Shen, Zechun Liu, and Marios Savvides. Attentive CutMix: An enhanced data augmentation approach for deep learning based image classification. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 3642–3646, 2020.

[14] Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. RandAugment: Practical automated data augmentation with a reduced search space. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[15] Jun Wang, Xiaohan Yu, and Yongsheng Gao. Mask guided attention for fine-grained patchy image classification. In *IEEE International Conference on Image Processing (ICIP)*, 2021.

[16] Dmitry Demidov, Muhammad Hamza Sharif, Aliakbar Abdurahimov, Hisham Cholakkal, and Fahad Shahbaz Khan. SM-ViT: Salient mask-guided vision transformer for fine-grained classification. In *International Conference on Computer Vision Theory and Applications (VISAPP)*, 2023.

[17] Li Ren, Chen Chen, Liqiang Wang, and Kien Hua. DA-VPT: Semantic-guided visual prompt tuning for vision transformers. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4353–4363, 2025.

[18] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[19] Ross Wightman. PyTorch Image Models. https://github.com/huggingface/pytorch-image-models, 2019.

[20] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[21] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations (ICLR)*, 2017.

[22] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zach DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[23] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman and Hall/CRC, 1993.